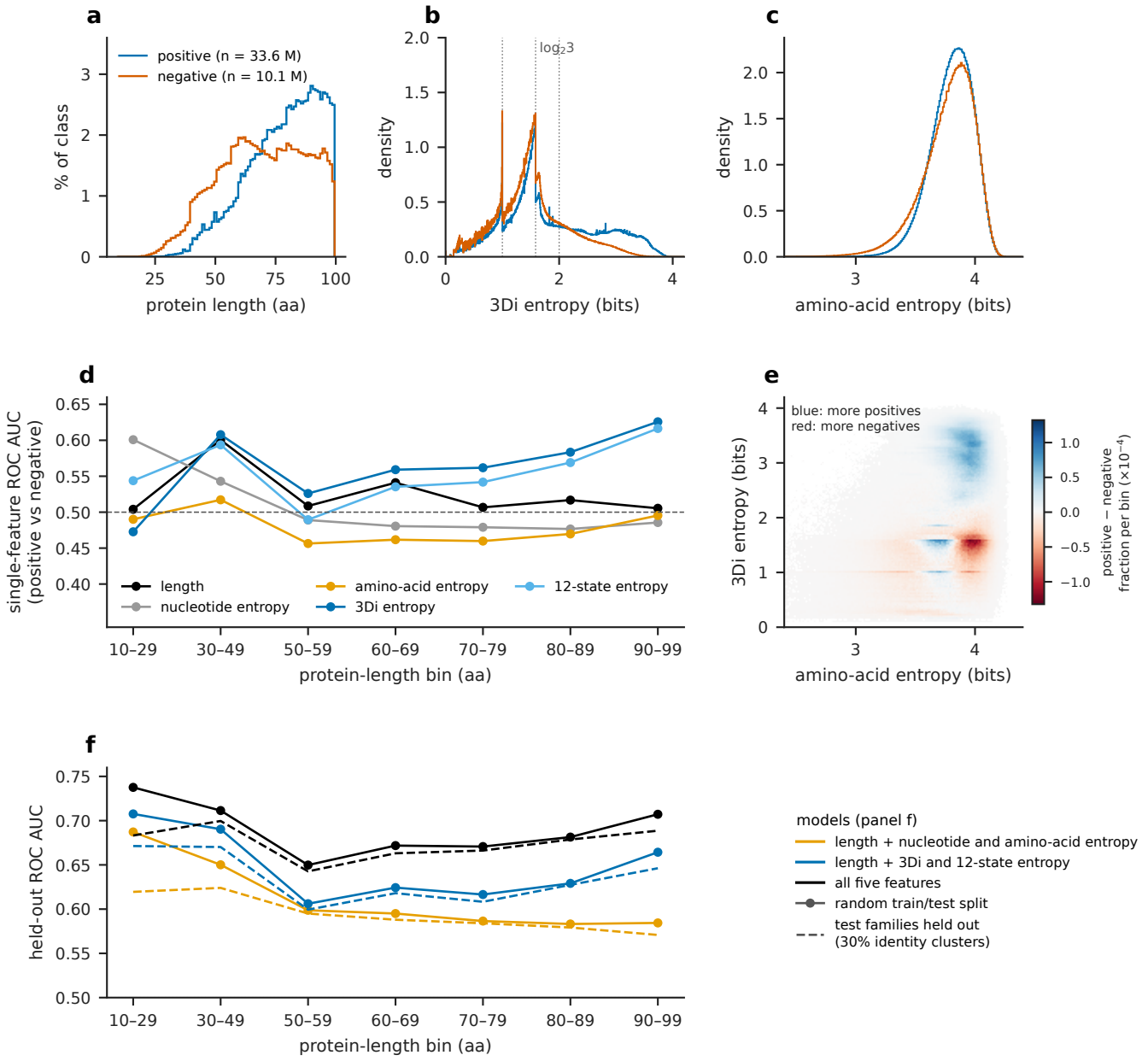

a
positive (n = 33.6 M)
negative (n = 10.1 M)
% of class
protein length (aa)
b
density
log2 3
3Di entropy (bits)
c
density
amino-acid entropy (bits)
d
single-feature ROC AUC
(positive vs negative)
length
nucleotide entropy
amino-acid entropy
3Di entropy
12-state entropy
protein-length bin (aa)
e
blue: more positives
red: more negatives
3Di entropy (bits)
amino-acid entropy (bits)
positive − negative
fraction per bin (×10⁻⁴)
f
held-out ROC AUC
protein-length bin (aa)
models (panel f)
length + nucleotide and amino-acid entropy
length + 3Di and 12-state entropy
all five features
random train/test split
test families held out
(30% identity clusters)